TS3000 series Robot Controller

TS3000 SCARA / LINEAR system
TS3100 SCARA / LINEAR / 6AXIS system
TSL3000 / TSL3000E SCARA system
TSL3100 / TSL3000E 6AXIS system
TSL3200E 6AXIS system

INSTRUCTION MANUAL

USER PARAMETER MANUAL

Rev.13 December 19, 2023

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

Copyright 2020 by SHIBAURA MACHINE CO.,LTD.
All rights reserved.

No part of this document may be reproduced in any form without obtaining prior written permission from SHIBAURA MACHINE CO.,LTD.

The information contained in this manual is subject to change without prior notice to effect improvements.

Preface

This manual describes the machine parameters for small-size SCARA robot controller "**TS3000 series**" and "**TSL3000 series**".
The machine parameters are registered in the controller under the text file named "**USER.PAR**". Like the SCOL program, they can be edited by the program editor.
For use of the program editor, see the Operator's Manual.

⚠ WARNING	
Prohibited	● Do not set values other than those written in the parameter manual. If you set values other than those written in the parameter manual, you will be struck by the robot with unexpected behavior. ● Do not delete a parameter file. If you delete a parameter file, you will be struck by the robot with unexpected behavior. The robot will operate with different settings from those before the deletion of the parameter file. ● Do not change the name of a parameter file. If you change the name of a parameter file, you will be struck by the robot with unexpected behavior. The robot will operate with different settings from those before the change of the parameter file name.

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damages to assets and to assure correct use.
Make sure that you well understand the following details (indications and symbols) before reading this manual.
Always observe the information that is noted.
To ensure safe work related to the robot from installation to operation, be sure to read the separate [Safety Manual] carefully before actually starting work.

[Explanation of indications]

Indication	Meaning of indication
⚠ DANGER	This means that "incorrect handling will lead to fatalities or danger of serious injury is imminent."
⚠ WARNING	This means that "incorrect handling will lead to fatalities or serious injuries."
⚠ CAUTION	This means that "incorrect handling may lead to personal injuries [1)]."
NOTICE	This means that "incorrect handling may lead to physical damage [2)]" or indicates advice on how to proceed with the work.

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to extended damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words near the symbol.
	This means that the action is instruction (must be done). Details of the instruction are indicated with pictures or words near the symbol.
	This means caution/warning (what to be aware of). Details of the caution are indicated with pictures or words near the symbol.

Table of Contents

USER PARAMETER MANUAL 1
PREFACE 3
[U00] EXTERNAL MODE FULL CONTROL 7
[U01] SYSTEM BASE COORDINATE 7
[U02] NOWAIT MODE FOR MULTI TASK 8
[U03] DEFAULT MOVESYNC MODE 9
[U04] TESTRUN MODE OVERRIDE 9
[U05] OVERRIDE ON SLOW SPEED SIGNAL 10
[U06] SERIAL PORT SETTING 11
[U07] SPECIFY SIGNAL FOR EXTSELECT 12
[U08] RESERVE 13
[U09] AUXILIARY SIGNAL 13
[U10] DEFAULT TEACHING MODE 15
[U11] I/O MODE 15
[U12] EXTEND I/O SETTING 16
[U13] SEQUENCE PARAMETER 16
[U14] SOFTWARE LIMIT 17
[U15] PLC ALARM 18
[U16] FIELD BUS 19
[U17] PASSWORD 21
[U18] ACCEL LIMIT FUNCTION 21
[U19] MASTER MODE 22
[U20] PLC COMMUNICATION REGISTER ADDRESS 22
[U21] SERIAL DATA ERROR CHECK 23
[U22] COMMON DATA & PROGRAM FILE FUNCTION SETUP 24
[U23] BYPASS FUNCTION PARAMETER 25
[U24] SPEED LIMIT FUNCTION 27
[U25] FUNCTION SELECT SWITCH 28
[U26] PAYLOAD FOR MANUAL OPERATION 34
[U27] STRINGIO.PAR RESPONSE COMMAND ADDITION SETTING 35
[U27] FORCE DETECTION LIMIT[N][NM] (6 AXES) 36
[U28] PASS COMMAND MODE SW 37
[U28] POSITION SHIFT LIMIT[MM][DEG] (6 AXES) 38
[U29] COMMUNICATION SIZE OF ALARM HISTORY 39
[U30] 3-AXIS ORIGIN CHANGE FUNCTION 40
[U31] JOG SPEED CHANGE FUNCTION 42
[U32] JOGGING SPEED[%] 43
[U33] INCHING SPEED[%] 44
[U34] INCHING AMOUNT [MM] [DEG] 45
[U35] SEQUENCE FUNCTION SELECT SWITCH 46
[U36] VIBRATION SUPPRESSION FUNCTION 49

[U37] ALARM FILE DUMP FUNCTION 50
[U38] ALARM FOR SIGNAL BY FILE OPERATION 52
[U39] WHEN THE OVERRIDE RETENTION FUNCTION POWER FAILURE 52
[U40] SMOOTH SPEED CONTROL FUNCTION 53
[U41] SMOOTH ALARM FUNCTION 53
[U42] INPUT ALARM RESTART FUNCTION 54
[U43] M-TO CONFIG ALARM FUNCTION 55
[U44] SIGNAL POINT DATA NAME CHANGE FUNCTION 56
[U45] FUNCTION WHICH TURNS ON AN ALARM SIGNAL TO ALL ALARMS 56
[U46] PULSE OUTPUT FUNCTION 57
[U47] PLC OVERRIDE LIMIT 59
[U48] NCBOY ADDITIONAL AXIS 60
[U49] NCBOY AXIS INCHING AMOUNT 61
[U50] SIGNAL FEED HOLD FUNCTION 62
[U51] 3-AXIS TORQUE DISPLAY (SCARA) 62
[U52] OUT OF LIMIT ALARM FUNCTION 63
[U53] TORQUE COMMAND ENABLE FLAG (6-AXIS) 64
[U54] COLLISION DETECTION FUNCTION 65
[U55] TOOL CALIBRATION FUNCTION 67
[U56] FUNCTION SELECT SWITCH2 70
EXTRNSEL.PAR SETTING PROGRAM FILE FOR FILE SELECT SIGNAL LINE 74
APPENDIX 1. USER PARAMETERS ADOPTION HISTORY 76

[U00] EXTERNAL MODE FULL CONTROL

Selection of control port

```
[U00] EXTERNAL MODE FULL CONTROL
{0:SIGNAL 1:RS232C 2:ETHERNET}
= 0
```

Set value: **= (EXTERNAL MODE execution form)**
Data type: Integer type
Data unit: None
Data range: 0 ~ 2
Number of data: 1
Description: Specify EXTERNAL MODE execution form.
0: External signal
1: Serial communication
2: Ethernet communication

[U01] SYSTEM BASE COORDINATE

BASE coordinate

```
[U01] SYSTEM BASE COORDINATE
= 0.0  0.0  0.0  0.0  0.0  0.0
```

Set value: **TS3000 = (X-axis BASE coordinate) (Y axis ...)(Z axis ...) (C axis ...) (Unused) (Unused)**
TS3000 (6-axis) = (X-axis BASE coordinate) (Y axis ...) (Z axis ...) (A axis ...) (B axis ...) (C axis ...)
Data type: Real number type
Data unit: None
Data range:
Number of data: 6
Description: Specify values in the BASE coordinate that are valid when the power supply is turned on.
Specify the values in the order that follows:
X Y Z C 0.0 0.0 for TS3000;
X Y Z A B C for TS3000 (6-axis)
Specify **0.0 0.0 0.0 0.0 0.0 0.0** when you wish to match the BASE coordinate system and the WORLD coordinate system.

⚠ WARNING	
Instruction	● If this parameter is changed after teaching operation, check the teach point. If you change this parameter after teaching operation, you will be struck by the robot with unexpected behavior.

[U02] NOWAIT mode for multi task

NOWAIT specified for multitasks

[U02]NOWAIT MODE FOR MULTI TASK (0:COMMON, 1:INDIVIDUAL) = 0

Set values **= 1 Independent control for each task (NOWAIT improved type) (SHIBAURA MACHINE's recommendation)**

= 0 Common use by tasks (NOWAIT previous type)

Data type: Integer type

Data unit: None

Data range: 0 or 1

Description: Specify a multitasking operation mode.
Specify whether system variable "NOWAIT" is controlled independently for each task or commonly used by tasks during multitasking operation. However, when the "NOWAIT" is commonly used by tasks, the SCOL programming becomes complicated. It is recommended to adopt a mode for setting independently for each task.

[U03] DEFAULT MOVESYNC MODE

Default of MOVESYNC

[U03] DEFAULT MOVESYNC MODE (0:ENABLE, 1:DISABLE) = 1

Set values	**= 0 Motion command synchronous mode (ENABLE MOVESYNC)** **= 1 Motion command asynchronous mode (DISABLE MOVESYNC)**
Data type:	Integer type
Data unit:	None
Data range:	0 or 1
Number of data:	1
Description:	Specify the synchronous or asynchronous mode for a motion command. Specify selection of the motion command synchronous mode that the system executes just before the next motion command and waits until positioning is completed during programmed operation or selection of the motion command asynchronous mode that the system pre-read and executes up to four (4) motion commands. In the motion command asynchronous mode, pass motion cannot be performed even if "PASS" is set to ENABLE.

[U04] TESTRUN MODE OVERRIDE

Teaching mode override

[U04] TEACHING MODE OVERRIDE = 20

Set value:	**Example 20 (which is a value limited to 20% of the maximum speed)**
Data type:	Integer type
Data unit:	%
Data range:	1 ~ 100
Number of data:	1
Description:	Specify the maximum speed of motion in teaching mode. *If 0 is set, operation is performed at the internal parameter speed that was set at shipping. *A value of 20 or more can be set, but the override is limited to 20% or less when operating in the teaching mode.

[U05] Override on slow speed signal

Override slow-speed command

[U05] OVERRIDE ON SLOW SPEED SIGNAL (0 - 100)[%] = 25

Set value:	**Example 25 (which is a value limited to 25% of the maximum speed)**
Data type:	Integer type
Data unit:	%
Data range:	1 ~ 100
Number of data:	1
Description:	Specify the speed at the command of slow speed (system input signal). Specify robot operating speed at the input of a slow-speed command signal. When a slow-speed command signal is input during automatic operation, specify at how many percent of the maximum speed the robot operates. *If 0 is set, operation is performed at the internal parameter speed that was set at shipping.

[U06] Serial port setting

Serial port

Specify the communication port (i.e., COM port).

```
[U06] SERIAL PORT SETTING
{Speed }(38400, 19200, 9600, 4800, 2400, 1200)
{Character}(7, 8)
{Parity }(0:Without, 1:Odd, 2:Even)
{Stop bit }(1, 2)
{COM1}
= 9600   8   0   1
{HOST}
= 9600   8   0   1
```

Set value: **{COM1}**
=(Speed) (Character length) (Parity) (Stop bit length) ←Setting of COM1 port
{HOST}
=(Speed) (Character length) (Parity) (Stop bit length) ←Setting of HOST port

Data type:

Data unit:

Data range:

Number of data: 2*4

Description: (Speed):Specify data communication speed. Select among six (6) speeds that follow.

38400 :38400 bps
19200 :19200 bps
9600 : 9600 bps
4800 : 4800 bps
2400 : 2400 bps
1200 : 1200 bps

(Character length):Specify the character length to be transferred.

8: 8 bits
7: 7 bits

(Parity):Specify the parity of characters to be transferred.

0: Without parity
1: Odd parity
2: Even parity

(Stop bit length):Specify the stop bit length of characters to be transferred.

1: Stop bit 1
2: Stop bit 2

*For details, see the Communication Manual.

[U07] Specify signal for EXTSELECT

Setting of file select signal line.

```
[U07] SPECIFY SIGNAL FOR EXTSELECT
{Signal No }(1 - )
{Bit length }(1 - 8)
= 101   4
```

Set value: Example **101 4** Four (4) extended input signals numbered 101 ~ 104.

Data type: Integer type

Data unit:

Data range: At least eight (8) successive signals from standard inputs 1 ~ 64, extended inputs 101 ~ 164, or field bus inputs 401 ~ 464.

Number of data: 2

Description: Set file selected signal line.

Specify the input signal from outside the controller to select an execution file, using the input signal.

Specify up to eight (8) consecutive input signals from standard inputs 1 ~ 64, extended inputs 101 ~ 164 or field bus inputs 401 ~ 464. To be more specific, specify the leading number of input signals to be used and the number of signals to be used by discriminating them by space.

= Specify in order (Signal No.), (Bit length).

(Signal No.): Specify the leading number of input signals to be used.

Set value		
	1 ~ 64	(Standard input)
	101 ~ 164	(Extended input)
	301 ~ 364	(Field bus input)
	401 ~ 464	(Field bus input)

Select the number from the above inputs.

(Bit length): Specify the number of signals to be used.

Set value TS3000: 1~8

[U08] RESERVE

Reserve parameter

```
[U08] RESERVE
= 0
```

Description: Set the reserve parameter as "0" because it is not currently set.

[U09] Auxiliary signal

Set the auxiliary signal display (AUX)

```
[U09] Auxiliary signal
   {Type }(0:Single, 1:Double)
   {Signal No. }(1 - 24)
   {Signal Name}(Max 10character)
= 0   201   "Hand out 1"
= 0   202   "Hand out 2"
= 0   203   "Hand out 3"
= 0   204   "Hand out 4"
= 0     0   "Not   Use   "
= 0     1   "Dout 1      "
= 0     2   "Dout 2      "
= 0     3   "Dout 3      "
= 0     4   "Dout 4      "
= 0     5   "Dout 5      "
= 0     6   "Dout 6      "
= 0     7   "Dout 7      "
= 0     8   "Dout 8      "
= 0     9   "Dout 9      "
= 0    10   "Dout 10     "
= 0    11   "Dout 11     "
= 0    12   "Dout 12     "
= 0    13   "Dout 13     "
= 0    14   "Dout 14     "
= 0    15   "Dout 15     "
```

Set value: Example 1: Set value **= 0 201 "Hand out 1"**

Output signal 201 is registered as the single solenoid and the display name is **Hand out 1**.

Example 2: Set value = **11 "out 1"**

Output signals 1 and 2 are registered as the double solenoid and the display name is **out 1**.

Data type:

Data unit:

Data range:

Number of data: 3*20

Description: Specify the data on the auxiliary signal display so that the robot I/O can be set ON and OFF from the teach pendant.

= Specify in order **(Solenoid), (Signal No.), (Signal name).**

Output signals for twenty (20) contacts can be specified.

(Solenoid): When the double solenoid is connected with output signals, the output signals should be controlled exclusively. When "Double" is specified, two (2) successive output signals are controlled exclusively.

0: Single solenoid (normal output)

1: Double solenoid (Two (2) consecutive output signals are output exclusively.)

Caution: This is the setting for auxiliary signal operation through the teach pendant. Even if the double solenoid is selected, signals are not output exclusively at automatic operation. Exclusive output of signals should be specified in the program.

Even if the double solenoid is specified, all output signals are OFF at power ON.

(Signal No.): Specify the number of output signal to be used.

When "Double" is specified, specify the smaller number of two (2) consecutive output signals. Output signals that can be specified are as follows:

0			Not registered.
1	~	66	Standard output
101	~	164	Extended output
201	~	214	Hand output
301	~	364	Fieldbus output

(Signal name): Specify the signal name displayed.

Up to ten (10) characters can be set.

[U10] Default Teaching mode

Initial value of manual guide

```
[U10] Default Teaching mode
{Coordinate     =       0:  Joint, 1:   Tool, 2:   Work, 3:   World}
{Speed =        0:  Low, 1:   Mid, 2:   High}
{Jog mode       =       0:  Jog, 1:   Inching, 2:   Free}
= 3 0 0
```

Description: This parameter is used for old models, and cannot be used with TS3000 series controllers.

[U11] I/O MODE

I/O mode

```
[U11] I/O MODE
{Default/User}(0:Default, 1:User RAM, 2:User FLASH, 3:User backup RAM)
= 0
```

Set value: **= (I/O operation mode)**

Data type: Integer type

Data unit:

Data range: 0 ~ 3

Number of data: 1

Description: Specify the I/O operation mode.
I/Os specified in the program are calculated by sequence, which are then input or output. This parameter selects the storage area of such sequence program.
TS3000 allows to back up the RAM area also by battery.
Specify "1" for the parameter, which should be changed to "3" after the sequence debug. (Value 1 or 3 is saved in the same area. When "1" is specified, the sequence is cleared by power OFF/ON.)

[U12] Extend I/O setting

Extended I/Os

```
[U12] EXTEND I/O SETTING
{Use/Not Use} (0:Not Use, 1:Use )
{Not Use}
{Not Use}
= 0   0   0
= 0   0   0
```

Set value: **= (Use/Not Use) (Reserved) (Reserved)** ← Setting of distribution I/O node 0.
= (Use/Not Use) (Reserved) (Reserved) ← Setting of distribution I/O node 1.

Data type: Integer type

Data unit:

Data range: 0 or 1

Number of data: 3*2

Description: Set the extended I/Os.
For the extended I/Os, two (2) nodes (node 1 and node 2) can be connected.
Node 0 is set in the upper level, and node 1 in the lower level.
(Use/Not use): Specify the presence or absence of extended I/O.
0 : Absence of extended I/O
1 : Presence of extended I/O
(Reserved): Reserved for future system extension. Specify zero (0).
(Reserved): Reserved for future system extension. Specify zero (0).

[U13] Sequence Parameter

Sequence parameter

```
[U13] SEQUENCE PARAMETER (User I/O mode only)
= 0   0   0   0   0   0   0   0
```

Set value: **= (R510) (R511) (R512) (R513) (R514) (R515) (R516) (R517)**

Data type: Integer type

Data unit:

Data range: 0 ~ 1

Number of data: 8

Description: Set the values of internal display that can be utilized as the contact inputs in the user's created sequence program.
This parameter is effective only when "User" (user create sequence) is selected by [U11] I/O MODE.

***This function is available in TS1000 and TS2000 series, and is disabled in TS3000 series.**

[U14] Software limit

Software limit

[U14] SOFTWARE LIMIT [deg][mm]
{+ Direction}
= 116.0 141.0 121.0 361.0 0.000 0.000 0.000 0.000
{- Direction}
= -116.0 -141.0 -1.0 -361.0 0.000 0.000 0.000 0.000

{+ Direction}

Set value: TS3000 = **(1 axis + joint limit) (2 axis...) (3 axis...) (4 axis...) (5 axis...) (Reserved) (Reserved) (Reserved)**
TS3000 (6-axis) = (1 axis + joint limit) (2 axis...) (3 axis...)(4 axis...) (5 axis...) (6 axis...) (7 axis...) (8 axis...)

Data type: Real number type

Data unit: deg or mm

Data range:

Number of data: 8

Description: Specify the software limit values (JLIMT).
Setting of + (plus) joint limit (+ soft stroke limit). Values set in the joint limit setting change mode (JLIMIT) are saved.

{- Direction}

Set value: TS3000 = **(1 axis - joint limit) (2 axis...) (3 axis...) (4 axis...) (5 axis...) (Reserved) (Reserved) (Reserved)**
TS3000 (6-axis) = **(1 axis - joint limit) (2 axis...) (3 axis...) (4 axis...) (5 axis...) (6 axis...) (7 axis...) (8 axis...)**

Data type: Real number type

Data unit: deg or mm

Data range:

Number of data: 8

Description: Specify the software limit values (JLIMT).
Setting of - (minus) joint limit (- soft stroke limit). Values set in the joint limit setting change mode (JLIMT) are saved.

⚠ WARNING	
 Instruction	●After setting a software limit, be sure to check that the software limit alarms (4-42*, 4-43*) are correctly issued when the robot moves out of the range. If you fail to check after setting a software limit, you will be struck by the robot with unexpected behavior.

[U15] PLC ALARM

PLC alarms

[U15] PLC ALARM	
= "8-269 PLC alarm 01	"
= "8-270 PLC alarm 02	"
= "8-271 PLC alarm 03	"
= "8-272 PLC alarm 04	"
= "4-077 PLC alarm 05	"
= "4-078 PLC alarm 06	"
= "4-079 PLC alarm 07	"
= "4-080 PLC alarm 08	"
= "1-037 PLC alarm 09	"
= "1-038 PLC alarm 10	"
= "1-039 PLC alarm 11	"
= "1-040 PLC alarm 12	"
= "1-041 PLC alarm 13	"
= "1-042 PLC alarm 14	"
= "1-043 PLC alarm 15	"
= "1-044 PLC alarm 16	"

Set value: Example = "8-269 **PLCALM01**"

Data type:

Data unit:

Data range: Up to 32 alphanumeric characters

Number of data: 16

Description: Specify the PLC alarm message.

The PLC alarm is an alarm which can be generated by sequence when "User" is selected for "[U11] I/O mode". In this setting, message for each of such alarm can be specified.

Specify an alarm message to be displayed in the area flanked by the quotation marks ("). Though up to thirty-two (32) characters can be specified, the leading six (6) characters are used to identify the alarm number, and a total of twenty-six (26) characters can be used for the message. DO NOT change the leading six (6) characters (alarm code) which are used to identify the alarm number.

[U16] Field bus

Field bus

[U16] FIELDBUS
{type / Node Addr / Speed / Byte Order}
= -1 -1 -1 -1

Example: Set value = 37 4 1 0, (Type) (Node) (Transmission rate) (Type of master)

Data type: Integer type

Data unit:

Data range:

Number of data: 4

Description: Specify the field bus. The field bus is an optional function.

(Type): Specify the type of the field bus.

–1 : Without field bus option (initial value)

1 : PROFIBUS

3 : EtherNet/IP

4 : EtherCAT

5 : PROFINET

37 : DeviceNet

144 : CCLINK

(Node): Specify the node address of the field bus.

This address should not be identical with another device in the same network.

The setting range differs with the type of the field bus.

–1 : Without field bus option (initial value)

0 ~ 127 : PROFIBUS

0 ~ 63 : DeviceNet

1 ~ 64 : CCLINK

0 : EtherNet/IP

0 : EtherCAT

0 : PROFINET

(Transmission rate) : Specify the transmission rate.

Setting of PROFIBUS is unnecessary. Make it coincide with the master transmission rate.

Specify the transmission rate of DeviceNet.

0 : 125 kbps

1 : 250 kbps

2 : 500 kbps

Specify the transmission rate of CCLINK.

0 : 156 kbps

1 : 625 kbps

2 : 2.5 Mbps

3 : 5 Mbps

4 : 10 Mbps

This is not used for EtherNet/IP, EtherCAT, or PROFINET, so set to 0.

(Type of master) : Specify the type of the field bus master.

The bit string (endian) differs with the master. Make the value identical with the master.

0 : 16-bit big endian (DRM21, DeviceNet, PROFIBUS made by OMRON)

(A1SJH, CCLINK, PROFIBUS made by MELSEC)

1 : 16-bit little endian (S7 315DP2, PROFIBUS made by SIEMENS)

(Keyence KV-5500,EtherNet/IP)

(NJ series, EtherCAT made by OMRON)

(SIMATIC S7-1200, PROFINET made by SIEMENS)

2 : 32-bit big endian

3 : 32-bit little endian

*For detailed settings of EtherNet/IP, EtherCAT, PROFINET, and DeviceNet communications, FIELDBUS.PAR needs to be set.

*For details, see the Fieldbus Slave Manual.

[U17] Password

Password

[U17] Password = " "

Set value: **= (Password up to 8 characters)**
Data type:
Data unit:
Data range:
Number of data: 1
Description: The password is a character string when using the password function.
The password function is optional.

[U18] Accel limit function

Function for limiting acceleration and deceleration according to Z -axis height

[U18] ACCEL LIMIT FUNCTION = 0

Set value: **= (Function effective/ineffective)**
Data type: Integer type
Data unit:
Data range: 0 or 1
Number of data: 1
Description: When the parameter is made valid, acceleration/deceleration is limited according to Z-axis height. This function is effective only for the SCARA robot system.

0 : Ineffective function
1 : Effective function

[U19] Master mode

Master mode

```
[U19] MASTER MODE
= "EXT.SIGNAL"
```

Set value: **= (Specify master mode after the power supply is turned on.)**
Data type:
Data unit:
Data range:
Number of data: 1
Description: This function is ineffective for TS3000 series. The master mode is set using the key switch and user parameter [U00]

[U20] PLC communication register address

PLC communication register address

```
[U20] PLC COMMUNICATION REGISTER ADDRESS
{Current Position/ Alarm/ Master Mode/ Alarm Resetting/ spare/ spare}
=      4    2560    2580       0       0       0
```

Set value: **See the description below.**
Data type: Integer type
Data unit:
Data range:
Number of data: 6
Description: This parameter is exclusively used for the die-cast unloading robot.
Communication is done with the TC200 made by SHIBAURA MACHINE.
Current Position : Address where the current coordinate of the robot (in the work coordinate system) is to be sent.
Alarm : Address where alarm information is to be sent.
Master Mode : Address where master mode information is to be sent.
Alarm Resetting : Address where alarm reset from the PLC is to be monitored.
Spare : Reserved
spare : Reserved

[U21] Serial data error check

Serial data error check

[U21] SERIAL DATA ERROR CHECK { 0:checked, 1:passed } = 0

Set value: **= (Specify whether there is presence or absence of error check for non-procedural communication and simple procedural communication.)**

Data type: Integer type

Data unit:

Data range: 0 or 1

Number of data: 1

Description: This parameter is used as the ON/OFF changeover flag of error check for non-procedural communication and simple procedural communication. Specify zero (0) normally.

0 : Error checking is carried out for non-procedural communication and simple procedural communication.

1 : Error checking is not carried out for non-procedural communication and simple procedural communication.

[U22] Common data & program file function setup

Common data and program function setup

```
[U22] COMMON DATA & PROGRAM FILE FUNCTION SETUP
{ 0:Disable, 1:DATA.TBL, 2:SCOL.PRG }
= 0
```

Set value: **= (Specify the ON/OFF flag of the common program and common data functions in the SCOL language specifications.)**

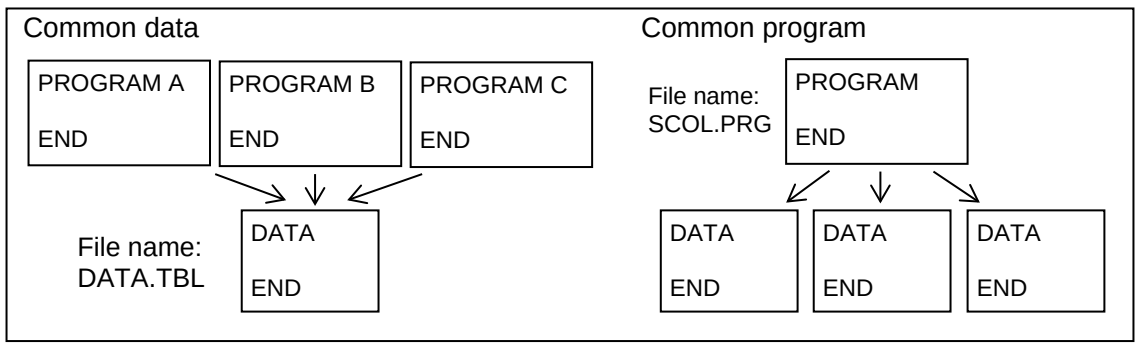
Data type: Integer type

Data unit:

Data range: 0 ~ 2

Number of data: 1

Description: This parameter serves as the ON/OFF flag of the common program and common data functions in the SCOL language specifications.
When the common data function is performed, the data (files containing teach point data) become common to multiple programs (from PROGRAM to END).
When the common program function is performed, multiple data are held for one common program. The name of the data when the common data function is performed is fixed to DATA.TBL, and the name of the program when the common program function is performed is fixed to SCOL.PRG.
0 : Disable
1 : Common data (DATA. TBL)
2 : Common program (SCOL. PRG)



[U23] Bypass function parameter

Bypass operation designation parameter

```
[U23] BYPASS FUNCTION PARAMETER
{ Function enable SW (0:Disable, 1:M-to & Bypass, 2:Bypass }
= 0
{ Retract position }
= 0.0     0.0     0.0     0.0     0.0     0.0     0.0     0.0
{ Z Axis offset parameter }
= 0.0     0.0
```

Set value: This function is used only for the SCARA robot system and LCDR robot system. The parameter should be set at zero (0) in other robot system. For details of setting, see the description below.

Data type: Real number type (Function enable SW is integer type.)

Data unit:

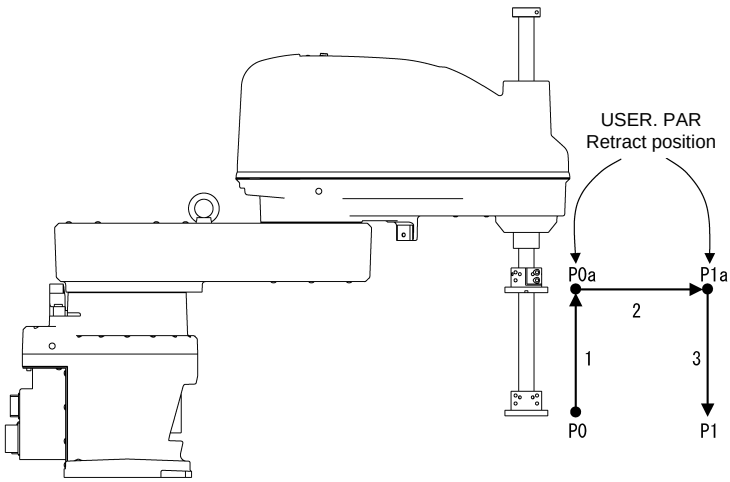
Data range:

Number of data: 11

Description: In the M–TO (move to teach point) function, as the robot arm moves directly to the teach point, it may collide with a workpiece present midway.
When this bypass (move to teach point by bypassing) function is made effective, the robot once retracts its hand, moves to just before the target position, then to the target position.

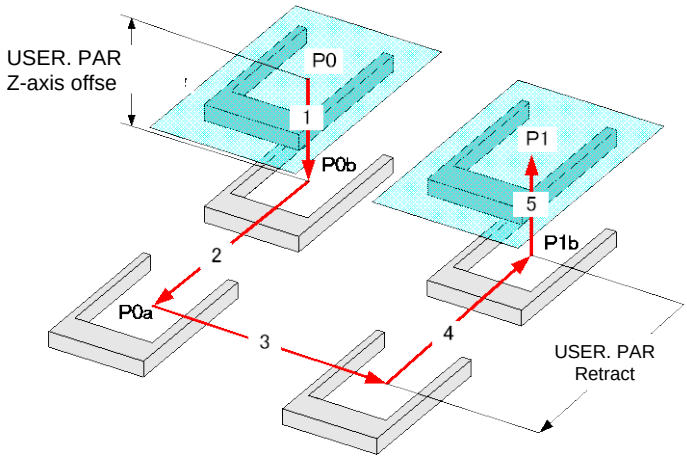
<SCARA robot>

1. Moves by retracting from current position P0 to P0a.
2. "Parallel move + rotation" from position P0a to P1a.
3. Moves by extending from position P1a to P1.



<LCDR robot>

1. Leaves a workpiece.
2. Moves by retracting from current position P0 to P0a.
3. "Parallel move + rotation" from position P0a to P1a.
4. Moves by extending from position P1a to P1.
5. Contacts the workpiece.



(1) Bypass function ON/OFF setting

= 2 (0 : OFF, 1: M–TO & bypass ON, 2: Only bypass ON)

(2) Retract position parameter (Specify the absolute position rather than the relative travel distance.)

= 0.0 0.0 **200.0** 0.0 0.0 0.0 0.0 0.0 (For SCARA robot: Only Z axis is effective.)

↑ Z-axis

= **100.0** **100.0** 0.0 0.0 0.0 0.0 (For LCDR robot: Only RR axis and RL axis are effective.)

↑ RR-axis ↑ RL-axis

Z-axis offset parameter for LCDR

= –10.0 –10.0

⚠ WARNING	
Instruction	● Make sure that the target position is correctly set before performing teach point movement. If the target position is not correctly set before performing teach point movement, you will be struck by the robot with unexpected behavior.

[U24] Speed limit Function

Maximum operation speed limit

[U24] SPEED LIMIT FUNCTION { 0:Disable, 1:Enable } = 0

Set value:	= (Specify operating speed limit.)
Data type:	Integer type
Data unit:	
Data range:	0 or 1
Number of data:	1
Description:	When this function is made valid, operating speed of the robot is controlled in such a manner that it will not exceed the maximum speed. 0 : Disable 1 : Enable

[U25] Function select switch

Selection of enabling or disabling controller functions

[U25] FUNCTION SELECT SWITCH
= 0 0 0 0 0 0
= 0 0 0 0 0 0
= 0 0 0 0 0 0
= 0 0 0 0 0 0

Set value:	**= (FUNCTION 1) (FUNCTION 2) (FUNCTION 3) (FUNCTION 4) (FUNCTION 5) (FUNCTION 6)** **= (FUNCTION 7) (FUNCTION 8) (FUNCTION 9) (FUNCTION10) (FUNCTION11) (FUNCTION12)** **= (FUNCTION13) (FUNCTION14) (FUNCTION15) (FUNCTION16) (FUNCTION17) (FUNCTION18)** **= (FUNCTION19) (FUNCTION20) (FUNCTION21) (FUNCTION22) (FUNCTION23) (FUNCTION24)**
Data type:	Integer type
Data unit:	
Data range:	0 or 1
Number of data:	6*4
Description:	This parameter selects ON or OFF of the controller functions. The functions which can be effective are assigned to each element. 1 : Function ON 0 : Function OFF

The effective functions of each element are described below.

Function	Item Name	Description	Setting Example	Supported Systems
(FUNCTION 1)	INPUT character string conversion function	In the INPUT process, this switches the process when a non-numerical character string is received. 0 : Function disabled. (Invalid Channel error occurs when a non-numerical character string is received.) 1 : Convert to 0 when a non-numerical character string is received. 2 : Convert to a user-selected number when a non-numerical character string is received. For details, see the Communication Manual, section 4.3.	[U25]Function switch = **1** 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	All systems
(FUNCTION 2)	Selection of movement method for teaching point movement (MOV-TO)	This switches the operation command in the teaching point movement based on the guidance coordinate status. BYPASS teaching point movement is also identical. 0 : All MOVE operations 1 : When guidance coordinates are JOINT → MOVE operation When guidance coordinates are TOOL, WORK, or WORLD → MOVES (linear interpolation) operation	[U25]Function switch = 0 **1** 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	All systems
(FUNCTION 3)	TCmini latch register retain function	In [U11] I/O MODE 2:USER FLASH mode, the register setting is not retained in the same way as 3:USER BACKUP RAM mode. The register retain function in 2:USER FLASH mode is switched by switching the parameters. 0 : Not retained (Retained in the same way as before when [U11] is 3) 1 : Retained (Not retained when [U11] is 0 or 1)	[U25]Function switch = 0 0 **1** 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	All systems
(FUNCTION 4)	Output change in I/O and AUX screens	I/O changing from UTILITY → I/O and from UTILITY → AUX screen is allowed in TEACHING MODE only. 0 : Conventional support (changing allowed) 1 : Editing prohibited (changing allowed in TEACHING mode only)	[U25]Function switch = 0 0 0 **1** 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	All systems
(FUNCTION 5)	Reserved	0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 **1** 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION 6)	Reserved	0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 **1** = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	

*Note that even if the functions are identical to the TS2000, the FUNCTION numbers may differ for certain functions.

Function	Item Name	Description	Setting Example	Supported Systems
(FUNCTION 7)	8-280 Buzzer and fault signal	This is a function that turns off the buzzer sound by an external signal and that turns off the alarm for the CT3 safety relay operation. When ”8-280 Safety SW relay ON” is generated, 0 : Output is performed 1 : Output is not performed (No alarm output or buzzer output is performed.)	[U25]Function switch = 0 0 0 0 0 0 = **1** 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	All systems
(FUNCTION 8)	8-281 Buzzer and fault signal	This is a function that turns off the buzzer sound by an external signal and that turns off the alarm for the CT3 safety relay operation. When ”8-281 Emergency SW relay ON” is generated, 0 : Output is performed 1 : Output is not performed (No alarm output or buzzer output is performed.)	[U25]Function switch = 0 0 0 0 0 0 = 0 **1** 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	All systems
(FUNCTION 9)	8-014 Alarm istory setting	Specifies whether “8-014 Emergency Stop SW ON” is recorded on the alarm history. 0 : Recorded on the alarm history (default) 1 : Not recorded on the alarm history	[U25]Function switch = 0 0 0 0 0 0 = 0 0 **1** 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION10)	8-017 Alarm istory setting	Specifies whether “8-017 Safety SW ON” is recorded on the alarm history. 0 : Recorded on the alarm history (default) 1 : Not recorded on the alarm history	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 **1** 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION11)	Function to disable acceleration suppressing	This function disables acceleration suppressing when the SPEED command changes the motion speed. With this function enabled, the acceleration time is proportional to the speed, and thus it can be shortened at a lower speed. 0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 **1** 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA
(FUNCTION12)	Function to read and discard INPUT reception data other than during program execution	This function reads and disables INPU reception data and empties the buffer other than during program execution. (This function also clears data that has accumulated in the buffer during program execution when the program stops.) This function becomes enabled for COM1, IP1, IP2 and default. 0 : The communication buffer is cleared while the program stops. 1 : The communication buffer is not cleared while the program stops (if it is returned to the old specification).	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 **1** = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA

*Note that even if the functions are identical to the TS2000, the FUNCTION numbers may differ for certain functions.

Function	Item Name	Description	Setting Example	Supported Systems
(FUNCTION 13)	Speed limit function by Z-axis area	This suppresses acceleration in the Z-axis area set by the parameter. 0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = **1** 0 0 0 0 0 = 0 0 0 0 0 0	Cartesian
(FUNCTION 14)	PRINT output zero suppress cancel function	The PRINT command outputs 12-character fixed-length data. Data is right-aligned and the rest is padded with spaces. When this function is enabled, the spaces are not displayed, and the data is left-justified. 0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 **1** 0 0 0 0 = 0 0 0 0 0 0	All systems
(FUNCTION 15)	Special function for specific customers : I/O specify number during distribution	This turns on the specified DOUT bit during distribution (during robot operation). Also, after distribution is completed, the specified DOUT bit is turned off. 0 : Disabled Value other than 0 : Enabled (Setting value 1 to 64, 101 to 164, 301 to 364) If the same number as FUNCTION16 is specified, the function is disabled.	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 **1** 0 0 0 = 0 0 0 0 0 0	SCARA Cartesian
(FUNCTION 16)	Special function for specific customers : I/O specify number during output	This turns on the specified DOUT bit when the robot controller and PC are connected by COM1 and data is being output to COM1. Also, after output is completed, the specified DOUT bit is turned off. 0 : Disabled Value other than 0 : Enabled (Setting value 1 to 64, 101 to 164, 301 to 364) If the same number as FUNCTION15 is specified, the function is disabled.	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 **1** 0 0 = 0 0 0 0 0 0	SCARA Cartesian
(FUNCTION 17)	Special function for specific customers : Execution command (character string) conversion function from external device	0 : Disabled 1 : Enabled (Once the execution command that was set in STRINGIO.PAR is sent from an external device to the robot controller by COM1, this function turns on the DIN information output that is set in STRINGIO.PAR or DOUT for a fixed period of time.)	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 **1** 0 = 0 0 0 0 0 0	

*Note that even if the functions are identical to the TS2000, the FUNCTION numbers may differ for certain functions.

Function	Item Name	Description	Setting Example	Supported Systems
(FUNCTION 18)	Substituting 0 into insufficient sections of INPUT data	When waiting for input data at INPUT COM1, if the number of loaded data is less than the number of specified variables, a wait is performed for data input that was not received. When this function is enabled, the input wait status is canceled, and the missing data is filled by 0. 0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 **1** = 0 0 0 0 0 0	SCARA Cartesian
(FUNCTION 19)	Tool flange distance display function	This adds the tool flange distance to the WORLD and WORK screens of the POS screen of the UTILITY. The tool flange distance is displayed only when this function is enabled. 0 : Disabled 1 : Enabled *Function available in TS3000 only.	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = **1** 0 0 0 0 0	Wafer only
(FUNCTION 20)	Reserved	0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 **1** 0 0 0 0	
(FUNCTION 21)	Reserved	0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 **1** 0 0 0	
(FUNCTION22)	Support for lowercase in PRINT command	When this function is enabled, output by the PRINT command is output unchanged in lowercase. (Previously, lowercase was converted to uppercase when output by the PRINT command.) Example: (Previously): PRINT “test”,CR → "TEST" is output (Function enabled): PRINT “test”,CR → "test" is output 0 : Disabled 1 : Enabled (Data sent and received by TSPC is not converted to uppercase.)	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 **1** 0 0	All systems
(FUNCTION23)	MOVE-MOVES PASS function	When this function is enabled, MOVE and other joint interpolation commands and MOVES/C and other linear/circular interpolation commands can be connected by PASS. (Except for MOVEJ and DELAY, this is enabled in the PASS section for all operation commands.) 0 : Disabled 1 : Enabled	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 **1** 0	SCARA 6-axis

*Note that even if the functions are identical to the TS2000, the FUNCTION numbers may differ for certain functions.

Function	Item Name	Description	Setting Example	Supported Systems
(FUNCTION24)	Buzzer sound when alarm generated	Because the TSL3000 does not have an operation panel, no method is available for stopping the buzzer except from the teach pendant. The buzzer sound can be disabled by enabling this function. The buzzer sound is enabled by default. 0 : Buzzer sound is on. 1 : Buzzer sound is off.	[U25]Function switch = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 **1**	All systems

*Note that even if the functions are identical to the TS2000, the FUNCTION numbers may differ for certain functions.

⚠ WARNING	
Instruction	● Pay attention to a change in the teach point movement route caused when FUNCTION 2 of this parameter is changed. If [U25] FUNCTION SELECT SWITCH is changed, you will be struck by the robot with unexpected behavior.

[U26] PAYLOAD FOR MANUAL OPERATION

Payload for manual operation (kg, mm)

[U26] PAYLOAD FOR MANUAL OPERATION
{PAYLOAD parameter for manual operation[Mass(kg),Offset(mm)]}
= 0.0 0.0 0.0 0.0 0.0 0.0 0.0

Setting: **= (MASS) (Gx) (Gy) (Gz) (Reserved) (Reserved) (Reserved)**

Data format: Real number type

Data units:

Data range:

Data count: 7

Description: This sets the PAYLOAD value during manual guidance.

(MASS) : <Mass> The mass of the load applied to the end of the robot is specified in kilogram units.

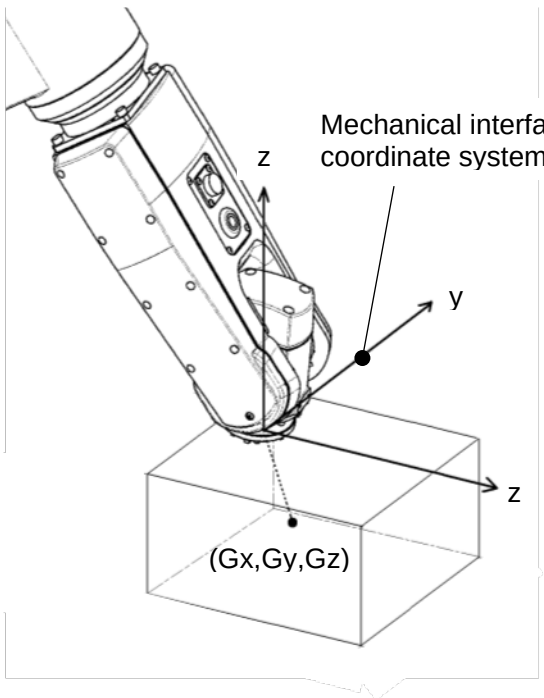
(Gx) to (Gz) : <Center of gravity offset> The distance from the center of gravity of the load applied to the end of the robot to the tool center at the tip is specified in millimeter units.

(Gx): <Center of gravity offset in X direction>

(Gy): <Center of gravity offset in Y direction>

(Gz): <Center of gravity offset in Z direction>

(Reserved): System-reserved parameter. Set to 0.



[U27] STRINGIO.PAR RESPONSE COMMAND ADDITION SETTING

STRINGIO.PAR response command setting (special function for wafer transfer system)

```
[U27] STRINGIO.PAR RESPONSE COMMAND ADDITION SETTING
{0:Both 1:command only 2:sign only}
= 0
{Reception response sign}
= "-"
{Processing completion sign}
= ">"
```

{0:Both 1:command only 2:sign only}

Setting:	**= (Response command type)**
Data format:	Integer type
Data units:	None
Data count:	1
Description:	This sets the response command type. 0 : Added symbol + command 1 : Command 2 : Added symbol

{Reception response sign}

Setting:	**= (Response command type)**
Data format:	Character type
Data units:	None
Data count:	1
Description:	This sets the response command addition data when receiving of the command is completed and the specified DOUT ON is completed. The setting response is 1 uppercase character.

{Processing completion sign}

Setting:	**= (Response command type)**
Data format:	Character type
Data units:	None
Data count:	1
Description:	This sets the response command addition data when sending of the data is completed and the specified DOUT OFF is completed. The setting response is 1 uppercase character.

*This function is enabled when FUNCTION 17 of [U25] FUNCTION SELECT SWITCH is enabled.
*For vertical articulated (6-axis) robot system, [U27] FORCE DETECTION LIMIT [N] [Nm] (force detection limit) in the next section is applied for this function.

[U27] FORCE DETECTION LIMIT[N][Nm] (6 axes)

Force detection limit

[U27] FORCE DETECTION LIMIT [N][Nm]
{+ Direction}
= 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
{- Direction}
= 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0

{+ Direction}

Setting: **= (X-axis) (Y-axis) (Z-axis) (A-axis) (B-axis) (C-axis) (U-axis) (V-axis)**

Data format: Real number type

Data units: N or N·m

Data range:

Data count: 8

Description: Set the positive limit of the external force detected in the compliance control.

{- Direction}

Setting: **= (X-axis) (Y-axis) (Z-axis) (A-axis) (B-axis) (C-axis) (U-axis) (V-axis)**

Data format: Real number type

Data units: N or N·m

Data range:

Data count: 8

Description: Set the negative limit of the external force detected in the compliance control.
To set the limit in the negative direction, add a minus sign.

To disable limit detection, set the positive- and negative-direction limits to "0.0".
If a detected external force exceeds the value specified by this parameter, a level 8 alarm occurs.
(Currently, only X-, Y-, and Z-axes are enabled.)

*This function is for the vertical articulated (6-axis) robot system. For systems other than vertical articulated robots, [U27] STRINGIO.PAR RESPONSE COMMAND ADDITION SETTING (STRINGIO.PAR response command) in the previous section is applied.

[U28] PASS COMMAND MODE SW

PASS operation mode switching function

[U28] PASS COMMAND MODE SW {0:PASS PRIORITY MODE 1:NO WAIT MODE} = 0

Setting: **= (Specifies the PASS operation mode)**

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 1

Description: This function is used to select whether to perform the PASS operation after completing PASS path generation or without completing PASS path generation.

0 : PASS operation is performed after path generation is completed.

*The PASS operation is performed, but if the program processing time is long between PASS operation commands, a WAIT may be generated before starting the PASS operation (for compatibility with previous versions).

1 : The PASS operation is performed without waiting for completion of PASS path generation.

*The PASS operation immediately after the WAIT MOTION≧100 command can no longer be performed.

*For vertical articulated (6-axis) robot system, [U28] POSITION SHIFT LIMIT [mm][deg] (position shift limit) in the next section is applied for this function.

[U28] POSITION SHIFT LIMIT[mm][deg] (6 axes)

Position shift limit

[U28] POSITION SHIFT LIMIT[mm][deg]
{+ Direction}
= 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
{- Direction}
= 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0

{+ Direction}

Setting	**= (X-axis) (Y-axis) (Z-axis) (A-axis) (B-axis) (C-axis) (U-axis) (V-axis)**
Data format:	Integer type
Data units:	mm or deg
Data range:	
Data count:	8
Description	Set the limit in the positive direction for position shift by the compliance control

{- Direction}

Setting	**= (X-axis) (Y-axis) (Z-axis) (A-axis) (B-axis) (C-axis) (U-axis) (V-axis)**
Data format:	Integer type
Data units:	mm or deg
Data range:	
Data count:	8
Description:	Set the limit in the negative direction for position shift by the compliance control. To disable limit detection, set the positive- and negative-direction limits to "0.0". If a detected value exceeds the value specified by this parameter, a level 8 alarm occurs. (Currently, only X-, Y-, and Z-axes are enabled.)

*This function is for the vertical articulated (6-axis) robot system. For systems other than vertical articulated robots, [U28] PASS COMMAND MODE SW (PASS operating mode switching function) in the previous section is applied.

[U29] COMMUNICATION SIZE OF ALARM HISTORY

Alarm history display switching function

```
[U29] COMMUNICATION SIZE OF ALARM HISTORY
{0:256 1:128}
= 0
```

Setting	**= (Specifying display of alarm history)**
Data format:	Integer type
Data units:	
Data range:	0 or 1
Data count:	1
Description:	This function is used to select whether the alarm history shows a maximum of 128 items or 256 items. *This function is enabled for TSPC version V1.64 or later, and for TSPC6ax version V1.53 or later. For TSPC earlier version than V1.64, and TSPC6ax eversion V1.53 or earlier, the number of alarms displayed in the alarm history is 128 at the maximum, regardless of this function. 0 : Alarm history shows a maximum of 256 items. 1 : Alarm history shows a maximum of 128 items.

[U30] 3-AXIS ORIGIN CHANGE FUNCTION

3-axis origin change function

[U30] 3-AXIS ORIGIN CHANGE FUNCTION
{0:Disabled 1:Enabled}
to 0
{RESERVE}
= 0.0

{0:Disabled 1:Enabled}

Setting: **= (Function enable/disable)**

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 1

Description: This changes the 3-axis origin position of the SCARA robot.

0 : Function disabled (bottom edge is 0)

1 : Function enabled (top edge is 0)

By default, the 3-axis bottom edge has a software limit of "0", but when this function is enabled, the 3-axis top edge is set to 0.

This function is enabled for SCARA robot systems only.

{RESERVE}

Setting:

Data format: Real number type

Data units:

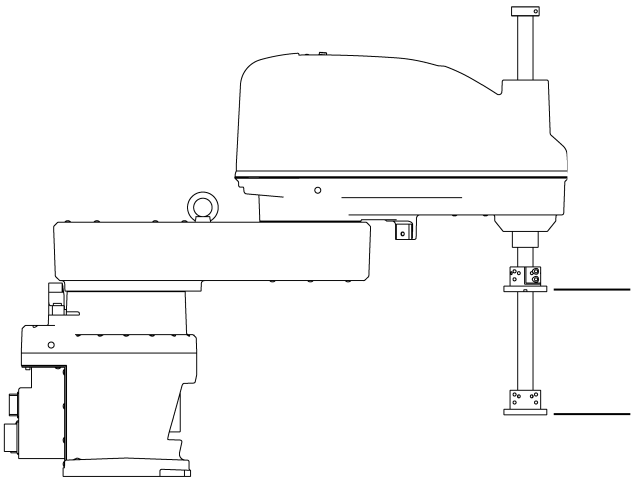
Data range:

Data count: 1

Description: System reserved.

Additional description

* For the position of the 3-axis top edge, the 3-axis origin is shifted using the [U14] 3-axis positive (+) software limit value (all numbers after the decimal point are rounded off).
The current position after inversion = the position before inversion – the [U14] 3-axes positive software limit value



[U14] Software limit setting value	[U30] 0:Disabled	[U30] 1:Enabled
200.5	200.5	0.5
	200.0	0.0
	0	-200.0
-0.5	-0.5	-200.5

Notes
This function shifts the origin based on the setting value for [U14]. If [U14] is changed, the Z-axis position also changes.

When this function is enabled ([U30] is set to 1: Enabled), all origin setting related functions are disabled. To use an origin setting related function, disable this function ([U30] is set to 0: Disabled).

Functions disabled when [U30] is set to 1 : Enabled
• SOFT LIMIT screen
• ZERO POSITION screen
• REORIG screen
• MKORG screen

[U31] JOG SPEED CHANGE FUNCTION

Jog speed change function

[U31] JOG SPEED CHANGE FUNCTION {0:Disabled 1:Enabled} = 0

Setting:	**= (Function enable/disable)**
Data format:	Integer type
Data units:	
Data range:	0 or 1
Data count:	1
Description:	This is a function that enables any user-selected setting for the jog speed, inching speed, and inching amount. 0 : Function disabled (Robot operates using the internal parameters that were set at shipping.) 1 : Function enabled (The job speed, inching speed, and inching amount that were set in user parameters [U32] to [U34] are applied.)

*Robot operation is performed at the same speed as before (internal parameters that were set at shipping) for teaching point movement and BYPASS, which change the jog speed.

⚠ WARNING	
 Instruction	●When this parameter is enabled, check the setting values of [U32], [U33], and [U34]. If [U31] JOG SPEED CHANGE FUNCTION is enabled, you will be struck by the robot with unexpected behavior.

[U32] JOGGING SPEED[%]

Jogging speed

[U32] JOGGING SPEED								
{Rectangular Coordinate [%]}								
=	10							
=	5							
=	1							
{Joint Coordinate [%]}								
=	8	8	5	10	10	0	0	0
=	5	5	3	5	5	0	0	0
=	1	1	1	1	1	0	0	0

{ Rectangular Coordinate }

Setting: **= (Jog speed: High)**
= (Jog speed: Medium)
= (Jog speed: Low)

Data format: Integer type

Data units: %

Data range: 1 to 100

Data count: 1*3 (High, Medium, Low)

Description: The movement speed when performing jog guidance in the world coordinate system, workpiece coordinate system, and tool coordinate system is defined as a percentage (%) value of the maximum speed.

*When set to 0, operation is performed at the internal parameter speed that was set at shipping.

{ Joint Coordinate }

Setting: **= (1-axis jog speed: High) (2-axis…) (3-axis…) (4-axis…) (5-axis…) (6-axis…) (7-axis…) (8-axis…)**
= (1-axis job speed: Low) (2-axis…) (3-axis…) (4-axis…) (5-axis…) (6-axis…) (7-axis…) (8-axis…)

Data format: Integer type

Data units: %

Data range: 1 to 100

Data count: 8*3 (High, Medium, Low)

Description: The movement speed when performing jogging by joint guidance is defined as a percentage (%) value of the maximum speed for each axis.

*When set to 0, operation is performed at the internal parameter speed that was set at shipping.

[U33] INCHING SPEED[%]

Inching speed

[U33] INCHING SPEED								
{Rectangular Coordinate [%]}								
=	10							
{Joint Coordinate [%]}								
=	10	10	10	10	10	0	0	0

{ Rectangular Coordinate }

Setting: **= (Inching speed)**

Data format: Integer type

Data units: %

Data range: 1 to 100

Data count: 1

Description: The movement speed when performing inching in the world coordinate system, workpiece coordinate system, and tool coordinate system is defined as a percentage (%) value of the maximum speed.

*When set to 0, operation is performed at the internal parameter speed that was set at shipping.

{ Joint Coordinate }

Setting: **= (1-axis inching speed) (2-axis…) (3-axis…) (4-axis…) (5-axis…) (6-axis…) (7-axis…) (8-axis…)**

Data format: Integer type

Data units: %

Data range: 1 to 100

Data count: 8

Description: The movement speed when performing inching by joint guidance is defined as a percentage (%) value of the maximum speed.

*When set to 0, operation is performed at the internal parameter speed that was set at shipping.

*Be very careful when first performing guiding operation after setting to a value greater than the default

[U34] INCHING AMOUNT [mm] [deg]

Inching amount

[U34] INCHING AMOUNT								
{Rectangular Coordinate [mm][deg]}								
=	1.0							
=	0.5							
=	0.25							
{Joint Coordinate [mm][deg]}								
[0.10, 0.10, 1.00, 0.10, 0.10, 0.0, 0.0, 0.0]								
=	0.05	0.05	0.50	0.05	0.05	0.0	0.0	0.0
=	0.01	0.01	0.10	0.01	0.01	0.0	0.0	0.0

{ Rectangular Coordinate }

Setting: **= (Inching amount: Large)**
= (Inching amount: Medium)
= (Inching amount: Small)

Data format: Real number type

Data units: mm or deg

Data range: 0.01 to limit value

Data count: 1*3 (Large, medium, small)

Description: This sets the movement amount when performing inching in the world coordinate system, workpiece coordinate system, and tool coordinate system.

*When set to 0, operation is performed at the internal parameter movement amount that was set at shipping.

{ Joint Coordinate }

Setting: **=(1-axis inching amount: Large) (2-axis…) (3-axis…) (4-axis…) (5-axis…) (6-axis…) (7-axis…) (8-axis…)**
= (1-axis inching amount: Medium) (2-axis…) (3-axis…) (4-axis…) (5-axis…) (6-axis…) (7-axis…) (8-axis…)
= (1-axis inching amount: Small) (2-axis…) (3-axis…) (4-axis…) (5-axis…) (6-axis…) (7-axis…) (8-axis…)

Data format: Real number type

Data units: mm or deg

Data range: 0.01 to limit value

Data count: 8*3 (Large, medium, small)

Description: This sets the inching amount when performing inching by joint guidance.

*When set to 0, operation is performed at the internal parameter movement amount that was set at shipping.
*Be very careful when first performing guiding operation after setting to a value greater than the default setting.

[U35] SEQUENCE FUNCTION SELECT SWITCH

Selection of enabling or disabling sequence function

[U35] SEQUENCE FUNCTION SELECT SWITCH
= 0 0 0 0 0 0 0 0

Setting: **= (FUNCTION 1) (FUNCTION 2) (FUNCTION 3) (FUNCTION 4) (FUNCTION 5) (FUNCTION 6) (FUNCTION 7) (FUNCTION 8)**

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 8

Description: This selects whether functions using the sequence are enabled or disabled.
The function enabled for each element is assigned.
1 : Function on
0 : Function off

The parameter function and signal assignments are shown in the table below.

[U35] (from left)	Signal name	Function	Description
FUNCTION 1	H118(SEQFSW1)	Fieldbus system signal input/output function	When this function is enabled, the system signal is assigned to the Fieldbus. 0 : Disabled 1 : Enabled
FUNCTION 2	H119(SEQFSW2)	Fieldbus Data communication function	When this function is enabled, data communication with an external sequencer (CC-Link, EtherNet/IP, EtherCAT, PROFINET connection) is possible without editing the ladder program of the internal PLC. 0 : Disabled 1 : Enabled
FUNCTION 3	H11A(SEQFSW3)	Reserved *1	
FUNCTION 4	H11B(SEQFSW4)	Reserved *1	
FUNCTION 5	H11C(SEQFSW5)	Reserved *1	
FUNCTION 6	H11D(SEQFSW6)	Reserved *1	
FUNCTION 7	H11E(SEQFSW7)	Reserved *1	
FUNCTION 8	H11F(SEQFSW8)	Reserved *1	

*1: Reserved for function expansion.

(FUNCTION 1) Fieldbus system signal input/output function

This function assigns system signal inputs and outputs to the fieldbus.

[U35] SEQUENCE FUNCTION SELECT SWITCH

= **1** 0 0 0 0 0 0 0

When the user parameter [U35] SEQUENCE FUNCTION SELECT SWITCH line 1, column 1 (FUNCTION 1) is set to the following value, the fieldbus assignments are switched.

0: Function OFF (System signals are input/output to/from the external operation input/output signal line (SYSTEM).)

1: Function ON (System signal input/output is assigned to the fieldbus. Input/output to/from the external operation input/output signal line (SYSTEM) is disabled.)

Fieldbus signal assignment

Input signal

Fieldbus input	When function OFF	When function ON
FIELDBUS_IN-113	DIN(449)	STROBE (strobe)
FIELDBUS_IN-114	DIN(450)	PRG_RST (program reset)
FIELDBUS_IN-115	DIN(451)	STEP_RST (step reset)
FIELDBUS_IN-116	DIN(452)	CYC_RST (cycle reset)
FIELDBUS_IN-117	DIN(453)	DO_RST (output signal reset)
FIELDBUS_IN-118	DIN(454)	ALM_RST (alarm reset)
FIELDBUS_IN-119	DIN(455)	RUN (run)
FIELDBUS_IN-120	DIN(456)	EX_SVON (external servo ON)
FIELDBUS_IN-121	DIN(457)	STOP (stop)
FIELDBUS_IN-122	DIN(458)	CYCLE (cycle operation mode)
FIELDBUS_IN-123	DIN(459)	LOW_SPD (low speed command)
FIELDBUS_IN-124	DIN(460)	BREAK (braking and stop)
FIELDBUS_IN-125	DIN(461)	SV_OFF (servo OFF)
FIELDBUS_IN-126	DIN(462)	BZ_RST (buzzer reset)

* For details of system signals, refer to the "Interface Manual".

Output signal

Fieldbus output	When function OFF	When function ON
FIELDBUS_OUT-113	DOUT(449)	EMG_ST (emergency stop)
FIELDBUS_OUT-114	DOUT(450)	SV_RDY (servo operation ready)
FIELDBUS_OUT-115	DOUT(451)	ACK (acknowledge)
FIELDBUS_OUT-116	DOUT(452)	TEACH (manual mode)
FIELDBUS_OUT-117	DOUT(453)	INT (internal mode)
FIELDBUS_OUT-118	DOUT(454)	EXT_SIG (external mode)
FIELDBUS_OUT-119	DOUT(455)	EXT_232C (external mode)
FIELDBUS_OUT-120	DOUT(456)	SYS_RDY (operation ready)
FIELDBUS_OUT-121	DOUT(457)	AUTO_RUN (automatic operation)
FIELDBUS_OUT-122	DOUT(458)	CYC_END (cycle end)
FIELDBUS_OUT-123	DOUT(459)	LOW_ST (low speed mode)
FIELDBUS_OUT-124	DOUT(460)	BT_ALM (battery alarm)
FIELDBUS_OUT-125	DOUT(461)	ALARM (failure)
FIELDBUS_OUT-126	DOUT(462)	EXT_ETHER (external mode)

* For details of system signals, refer to the "Interface Manual".

(FUNCTION 2) Fieldbus data communication function

When this function is enabled, data communication with an external sequencer (CC-Link, EtherNet/IP, EtherCAT, PROFINET connection) is possible without editing the ladder program of the internal PLC.

[U35] SEQUENCE FUNCTION SELECT SWITCH

= 0 1 0 0 0 0 0 0

When the user parameter [U35] SEQUENCE FUNCTION SELECT SWITCH line 1, column 2 (FUNCTION 2) is set to 1, this function is enabled.

When this function is enabled, the followings are set.

H10E: Command error signal DATA_ERR -> Assigned to Fieldbus output (FO111)

H10F: Response completion signal DATA_ACK -> Assigned to Fieldbus output (FO112)

G10F: Request trigger signal DATA_TRIG -> Assigned to robot program "DIN(448)"

D700 to D70F: FieldBUS data input register -> Data transferred to D460 to D46F (command set register DATA_CMD)

D740 to D74F: FieldBUS data output register -> Data received from D660 to D66F (command response register DATA_RESP)

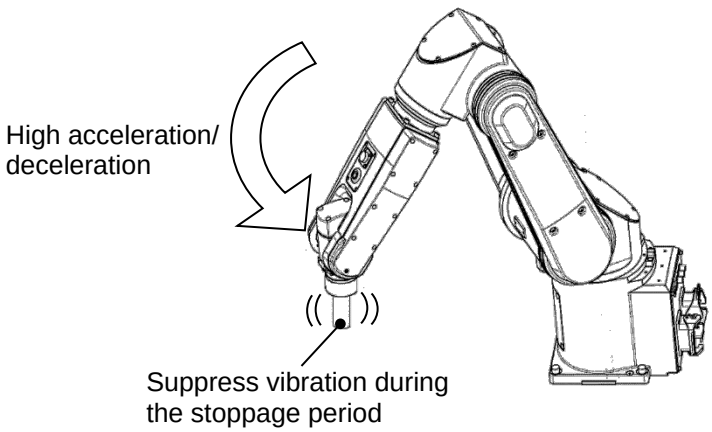
For details, see the Instruction Manual "Simplified PLC Functions" Chapter 12 Simplified PLC Data Communication.

[U36] VIBRATION SUPPRESSION FUNCTION

Vibration control function (available in TV800)

[U36] VIBRATION SUPPRESSION FUNCTION = 0

Setting:	**= (Enable or disable the function)**
Data format:	Integer type
Data units:	
Data range:	0 or 1
Data count:	1
Description:	This controls the arm vibration when the robot moves at a high acceleration. 1 : Function on 0 : Function off



Note that the vibration control function is disabled under the following conditions.

• When the program is stopped

• The payload command is never executed

• During the sensor-less compliance control

Once the servo is turned off, the vibration control is disabled until the above conditions are met.

*Note

When using this function, make sure to run the payload command in the robot program. If the payload command is not executed, this function does not work. Also, the payload parameters (mass, gravity center, etc.) must be set properly. Otherwise the robot may oscillate and stop.

•Error

If the robot oscillates due to the wrong setting of payload parameters, "008-055 Vibration Suppression Error" occurs. Once an error occurs, this function is disabled until the robot controller is turned OFF and ON.

* This function is available only in the vertical articulated (6-axis) robot TV800.

[U37] ALARM FILE DUMP FUNCTION

Alarm file dump function

```
[U37]ALARM FILE DUMP FUNCTION
{0:Disabled}
{1:Enabled(Enable function for all of the alarm)}
{2:Enabled(Latest alarm is enabled function except 8-014 or 8-017)}
{3:Enabled(Latest alarm is enabled function except 8-014)}
{4:Enabled(Latest alarm is enabled function except 8-017)}
= 0
```

Setting: **= (Enable or disable the function)**

Data format: Integer type

Data units:

Data range: 0 or 4

Data count: 1

Description: If this function is enabled, the ERROR.LOG file is created in the RAM of the robot controller when an alarm occurs.

0 : The function is disabled.

1 : The function is enabled for all alarms.

2 : The function is enabled when an alarm other than "8-014 Emergency Stop SW ON" or "8-017 Safety SW ON" is occurring.

3 : The function is enabled when an alarm other than "8-014 Emergency Stop SW ON" is occurring.

4 : The function is enabled when an alarm other than "8-017 Safety SW ON" is occurring.

Items of the ERROR.LOG file

Item	Description
Current alarm	Shows a maximum of 10 alarm numbers that are occurring.
Distribution state	Shows the allocation status (0: OFF, 1: ON, 2: WAIT).
Start position world	World coordinate system start position
Start position joint	Joint coordinate system start position
End position world	World coordinate system target position (the allocation status is ON) World coordinate system previous target position (the allocation status is not ON)
End position joint	Joint coordinate target position (the allocation status is ON) Joint coordinate previous target position (the allocation status is not ON)
Program execution line	The program execution line
Program execution status	The program execution status
Select the file name	The selected file name
[M01] CONTROLLER NO.	MACHINE.PAR [M01] controller number
[M02] MACHINE NO.	MACHINE.PAR [M02] controller number
File save time	Time on which the file is saved

[U38] ALARM FOR SIGNAL BY FILE OPERATION

Alarm in file operation via signals

[U38]ALARM FOR SIGNAL BY FILE OPERATION {0:ON 1:OFF} to 0

Setting:	**= (Alarm enabled/disabled)**
Data format:	Integer type
Data units:	
Data range:	0 or 1
Data count:	1
Description:	Enable/disable the alarm in file operation via signals 0 : Issues an alarm. 1 : Does not Issue an alarm. * The default is 0.

*For details about file operation via signals, see the Instruction Manual "Simplified PLC Functions" Chapter 6.

[U39] WHEN THE OVERRIDE RETENTION FUNCTION POWER FAILURE

Override retention function in case of power failure

[U39]WHEN THE OVERRIDE RETENTION FUNCTION POWER FAILURE {0:Disabled 1:Enabled} = 0

Setting:	**= (Enable or disable the function)**
Data format:	Integer type
Data units:	
Data range:	0 or 1
Data count:	1
Description:	When this function is enabled, the override is retained when power failure occurs. The override remains enabled when restarting the power supply. 0 : The function is disabled. (The override is initialized at power ON.) 1 : The function is enabled. (The override is not initialized at power ON. The override retained at power failure is enabled.)

[U40] SMOOTH SPEED CONTROL FUNCTION

SMOOTH speed control function

[U40]SMOOTH SPEED CONTROL FUNCTION
{0:Disabled 1:Enabled}
= 0

Setting: **= (Enable or disable the function)**

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 1

Description: When this function is enabled, the alarm "002-097 C Axis MOVE error (C-axis movement error)" does not occur.

0 : The function is disabled. (As per usual, the "002-097 C Axis MOVE error" occurs.)

1 : The function is enabled. (The "002-097 C Axis MOVE error" does not occur).

*This function is available only in vertical articulated (6-axis) robot systems.

[U41] SMOOTH ALARM FUNCTION

Function to disable the SMOOTH alarm

[U41]SMOOTH ALARM FUNCTION
{0:Function Disabled 1:001-025 & 001-026 Disabled 2:001-025 Disabled 3:001-026 Disabled}
= 0

Setting: **= (Alarm enabled/disabled)**

Data format: Integer type

Data units:

Data range: 0 or 3

Data count: 1

Description: This function enables/disables the SMOOTH related alarms "001-025 Over Deceleration (over deceleration)" and "001-026 SMOOTH impossible (SMOOTH command impossible)".

0 : As per usual, an alarm occurs.

1 : The "001-025 Over Deceleration" and "001-026 SMOOTH impossible" alarms do not occur.

2 : The "001-025 Over Deceleration" alarm does not occur.

3 : The "001-026 SMOOTH impossible" alarm does not occur.

* This function is available only in vertical articulated (6-axis) robot systems.

[U42] INPUT ALARM RESTART FUNCTION

Function to change INPUT error processing

```
[U42]INPUT ALARM RESTART FUNCTION
{0:Disabled 1:Enabled}
= 0
```

Setting: **= (Enable or disable the function)**

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 1

Description: When this function is enabled, an INPUT instruction error (002-046 Invalid Channel or 002-034 Ethernet connection error) occurs. If the program is re-executed after the program stops, the program is executed from the line next to the interrupted line. (Conventionally, if an error occurs to stop the program, when the error is corrected and then the program is re-executed, program reset occurs and the program is executed from the beginning.)

0 : The function is disabled. (As conventional, program reset occurs and the program is executed from the beginning.)

1 : The function is enabled. (When the program is re-executed, the program is executed from the line next to the interrupted line.)

[U43] M-TO CONFIG ALARM FUNCTION

Function to disable configuration alarm of teach point movement by linear interpolating operation

```
[U43]M-TO CONFIG ALARM FUNCTION
{0:Function Disabled 1:001-003 Disabled}
= 0
```

Setting: **= (Enable or disable the function)**

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 1

Description: When this function is enabled, the alarm “001-003 M-TO Config Err” is disabled.

(The alarm “001-003 M-TO Config Err” occurs if the configuration at completion of teach point movement by linear interpolating operation and the teach point configuration differ. If the targeted teach point configuration contains FREE, no alarm occurs regardless of the configuration after completion of teach point movement.)

0 : The function is disabled. (As conventional, “001-003 M-TO Config Err” occurs.)

1 : The function is enabled. (“001-003 M-TO Config Err” is disabled.)

*This function is available only in vertical articulated (6-axis) robot systems.

[U44] SIGNAL POINT DATA NAME CHANGE FUNCTION

Function to change signal point data name

```
[U44]SIGNAL POINT DATA NAME CHANGE FUNCTION
{0:PXXX 1:P(XXX)}
= 0
```

Setting: **= (Point data name setting)**
Data format: Integer type
Data units:
Data range: 0 or 1
Data count: 1
Description: When this function is enabled, the point data name during signal point data reading, teaching, teach point movement, and BAYPASS movement changes to the format of point data names P(1) to P(999) in the array.
The default format is P001 to P999
0 : As conventional, the format of point data name is P001 to P999.
1 : The point data name changes to the format of point data names P(1) to P(999) in the array.

[U45] FUNCTION WHICH TURNS ON AN ALARM SIGNAL TO ALL ALARMS

Function to change output alarm signal

```
[U45] FUNCTION WHICH TURNS ON AN ALARM SIGNAL TO ALL ALARMS
{0:Disabled 1:All Alarms 2:other than the emergency stop}
= 0
```

Setting: **= (Output alarm signal setting)**
Data format: Integer type
Data units:
Data range: 0 or 2
Data count: 1
Description: Conventionally, the output alarm signal is turned on when the 2-, 4- or 8-level alarm is turned on.
When this function is enabled, the condition under which the output alarm signal is turned ON can be changed.
0 : The output alarm signal is turned on for 2-, 4- or 8-level alarms as before.
1 : The output alarm signal is turned ON for all alarms.
2 : The output alarm signal is turned ON for non-emergency stop alarm.
*The default value is 0.

[U46] PULSE OUTPUT FUNCTION

Speed pulse output function

```
[U46] PULSE OUTPUT FUNCTION
{0:Disabled 1:Feedback mode 2:Command mode}
= 0
{Pulse Weight}
=1.0
{I/O Number For Trigger [1-64]}
=1
```

{0:Disabled 1:Feedback mode 2:Command mode}

Setting: **= (Pulse data type setting)**

Data format: Integer type

Data units:

Data range: 0 to 2

Data count: 1

Description: This function determines the type of output pulse data.
The {Feedback mode} calculates the robot end speed from the feedback value of the motor and outputs pulses. The {Command mode} outputs pulses based on the command speed to the robot. When the pulse output function is not used, set to {Disabled}.
*The default setting is {Disabled}.

{ Pulse Weight }

Setting: **= (Pulse weight per robot end movement amount setting**

Data format: Real number type

Data units: pulse/μm

Data range: 0.0 to limit value

Data count: 1

Description: This function calculates the pulse weight [pulse] per robot end movement amount [μm].
*Since the pulse output function cycle is 3 ms, actually this parameter is calculated as a number rounded down to a multiple of the calculation cycle 3 ms.

<Notes> The pulse output function is set to 1.0 pulse by default, but you can set to a larger value. In such a case, however, make the adjustment with the following points in mind.

■ When set to a smaller value

The calculation error of the end-of-robot speed can be absorbed.

■ When set to a larger value

The change of the pulse amount in relation to the robot end speed becomes more agile.

*The default setting is 1.0.

{ I/O Number For Trigger [1-64]}

Setting: **= (Trigger output setting)**

Data format: Integer type

Data units:

Data range: 1 to 64

Data count: 1

Description: In addition to the "TRIG" described in the "Interface Manual", a trigger output function is provided as standard.

Pins 5 and 6 of "PULOUT" (half-pitch 20-pin connector) located on the upper back of the TS3000/TS3100 robot controller are used for the trigger output (RS-422 differential output).

For conducting trigger output, use the "DOUT" command of the SCOL language.

*This is common to the conventional DOUT external output signal line, so be careful not to make duplicates when using this. Duplicate use may cause malfunctions.

*The default value is 1.

Controller system versions in which this function is available:

Main: X8GCAS-11F or higher (SCARA), X8GCAV-11K or higher (vertical articulated)

Servo: X8GLA-11A or higher (TS3000 series), X8YCB-11A or higher (TSL3000 series)

Controller main board: X8GLA-2 or higher (pulse output circuit added)

*For details of the pulse output function, refer to the "Additional Function Manual".

[U47] PLC OVERRIDE LIMIT

PLC override 0 setting function

[U47] PLC OVERRIDE LIMIT {Upper Limit} = 100 {Lower Limit} = 1

{Upper Limit}

Setting:	**= (Upper limit setting)**
Data format:	Integer type
Data units:	
Data range:	0 to 100
Data count:	1
Description:	This sets the PLC override upper limit. If set to a value higher than 100, this value is set to 100. *The default value is 100.

{Lower Limit}

Setting:	**= (Lower limit setting)**
Data format:	Integer type
Data units:	
Data range:	0 to 100
Data count:	1
Description:	This sets the PLC override lower limit. If set to a value lower than 0, this value is set to 0. *The default value is 1.

[U48] NCBOY ADDITIONAL AXIS

NCBOY additional axis function enabling/disabling selection

[U48] NCBOY ADDITIONAL AXIS {Use/Not Use } (0: Not Use, 1:Use) = 0 0

Setting: **= (NCBOY additional axis 1 enabling/disabling)**
(NCBOY additional axis 2 enabling/disabling)

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 2

Description: This selection enables/disables NCBOY additional axis functions.
*For SCARA robots, the number of additional axes that can be added is only one.

Controller system versions in which this function is available:

Main : X8GCAV-13F or higher (vertical articulated only)

Sequencer : X8GCC-06C or higher (TS3000 series), X8YCC-06C or higher (TS3000 series)

* For details of the NCBOY additional axis function, refer to the "Additional Function Manual".

[U49] NCBOY AXIS INCHING AMOUNT

NCBOY additional axis inching amount setting

[U49] NCBOY AXIS INCHING AMOUNT
{Joint Coordinate [mm][deg]}
= 1.0 1.0
= 0.5 0.5
= 0.1 0.1

Setting: **= (NCBOY additional axis 1 inching amount high)**
(NCBOY additional axis 2 inching amount high)
= (NCBOY additional axis 1 inching amount medium)
(NCBOY additional axis 2 inching amount medium)
= (NCBOY additional axis 1 inching amount low)
(NCBOY additional axis 2 inching amount low)

Data format: Real number type

Data units: mm or deg

Data range: 0.0001 to limit value

Data count: 2*3 (high, medium, low)

Description: This parameter specifies the inching amount in joint guiding operation of NCBOY additional axes from the teaching pendant of the robot controller.
*For SCARA robots, the number of additional axes that can be added is only one.

Controller system versions in which this function is available:

Main : X8GCAV-13F or higher (vertical articulated only)

Sequencer : X8GCC-06C or higher (TS3000series), X8YCC-06C or higher (TSL3000 series)

*For details of the NCBOY additional axis function, refer to the "Additional Function Manual".

[U50] SIGNAL FEED HOLD FUNCTION

Signal feed hold function

[U50] SIGNAL FEED HOLD FUNCTION {0:Disabled 1:Enabled} = 0

Setting: **= (Enabling/disabling function)**

Data format: Integer type

Data units:

Data range: 0 or 1

Data count: 1

Description: This parameter enables/disables the FEED HOLD function from the external signal.

When this function is set to enabled, FEED HOLD can be performed by the external signal.

FEED HOLD is set when the signal is OFF, and canceled ON.

* The default value is 0.

0 : Function disabled

1 : Function enabled

*For details of this function, refer to the Instruction Manual "Simplified PLC Functions".

[U51] 3-AXIS TORQUE DISPLAY (SCARA)

3-axis torque display change function (SCARA)

[U51] 3-AXIS TORQUE DISPLAY (SCARA) {0:[%] 1:[N] (Add Payload) 2:[N] (Not Add Payload)} = 0

Setting: **= (3-axis torque display setting)**

Data format: Integer type

Data units:

Data range: 0 to 2

Data count: 1

Description: This parameter changes the 3-axis torque display of "UTILITY" -> "MOTOR" on the teaching pendant between [%] and [N].

The default value is 0.

0 : Percent [%] (as before)

1 : Newton [N] (including PAYLOAD mass)

2 : Newton [N] (Not including PAYLOAD mass)

*This function is available only in SCARA robots.

[U52] OUT OF LIMIT ALARM FUNCTION

Out of limit error disabling function

[U52] OUT OF LIMIT ALARM FUNCTION {0:Disabled 1:Enabled} = 0

Setting: **= (Enabling/disabling function)**

Data format: Integer type

Data units:

Data range: 0 to 1

Data count: 1

Description: When enabled, this function disables the generation of "002-079 Point Out of limit" and "002-401 to 408 Axis* Out of limit" errors. (No error message will be issued).

Whether or not the above Out of limit error has occurred at the time of execution of a move command can be referred to using the robot language "OLTSTATUS". OLTSTATUS" is set to "1" if an Out of limit error has occurred, and is set to "0" if no Out of limit error has occurred at the time of the next move command execution.

0 : Function disabled ("002-079 Point Out of limit" and "002-401 to 408 Axis* Out of limit" errors are generated as before.)

1 : Function enabled (The generation of "002-079 Point Out of limit" and "002-401 to 408 Axis* Out of limit" errors are disabled.)

*This function is currently available only in SCARA robot systems.

Controller system versions in which this function is available:

Main :X8GCAS-14F or higher (SCARA robot system only)

[U53] TORQUE COMMAND ENABLE FLAG (6-AXIS)

TORQUE command enabling/disabling setting (6-axis)

[U53] TORQUE COMMAND ENABLE FLAG (6-AXIS) {0:Disabled 1: Axis 6 Enable 2:Axis 5,6 Enable 3 : Axis 4,5,6 Enable} = 0

Setting: **= (Enabling/disabling function)**

Data format: Integer type

Data units:

Data range: 0 to 3

Data count: 1

Description: When this function is enabled, the vertical articulated (6-axis) robot system can use a TORQUE command in the same way as the SCARA robot system.

However, the axis whose torque limit value is changed is set by this parameter. (Only for axes 4 to 6).

For details on how to use the TORQUE command, refer to the "Robot Language Manual (vertical articulated system)".

0 : Function disabled (The torque is not limited when the TORQUE command is used.)

1 :Function enabled for axis 6

2 :Function enabled for axes 5 and 6

3 :Function enabled for axes 4, 5, and 6

When a TORQUE command is executed with this parameter set to 0, "2-071 TORQUE Command Disable" occurs.

* This function is available only in vertical articulated (6-axis) robot systems.

* Controller system versions in which this function is available:

Main : X8GCAV-15A or higher (vertical articulated (6-axis) robot system only)

[U54] COLLISION DETECTION FUNCTION

Collision detection function (available in TV800)

```
[U54] COLLISION DETECTION FUNCTION
{0:Disabled 1:Enabled}
= 0
{Detection sensitivity[%]}(50 - 120)
= 100  100  100  100  100  100  100  100
```

{0: Disabled 1: Enabled}

Setting: **= (Enabling/disabling function)**

Data format: Integer type

Data units:

Data range: 0 to 1

Data count: 1

Description: This parameter enables/disables the collision detection function.

0: Function disabled

1: Function enabled

This function detects a collision and immediately stops the robot when it collides with peripheral equipment during teaching operation in the manual (TEACHING) mode, thereby reducing the load on the robot.

It is not guaranteed that no harm will be done to human workers.

In addition, this function can only reduce the load caused by a collision, and depending on the posture and speed at the time of collision, a part of the robot may be damaged.

This function is enabled when all of the following conditions are met.

- The model and version of controller are those in which this function is available.
- In Manual (TEACHING) mode.
- The compliance control function* is disabled.
- The active acceleration/deceleration function* is disabled.
- The parameter [U36] of the vibration suppression function is disabled.
- 1.5 seconds have elapsed since servo ON was completed.

{Detection sensitivity[%]}(50 - 120)

Setting: **= (Axis 1 detection sensitivity) (Axis 2 detection sensitivity) (Axis 3 detection sensitivity) (Axis 4 detection sensitivity) (Axis 5 detection sensitivity) (Axis 6 detection sensitivity) (Reserved) (Reserved)**

Data format: Integer type

Data units: %

Data range: 50 to 120

Data count: 8

Description: This parameter sets the sensitivity to detect collisions. Setting a higher value makes it easier to detect a collision, and a lower value harder. If a value outside the data range (50 to 120) is set, a parameter error will occur.

* Notes

- When you approach the robot for recovery after a collision detection, stop the surrounding equipment and ensure your safety before starting the work. If you approach the robot while peripheral equipment is still operating, you may come into contact with it, resulting in unexpected injury.

- After a collision detection, operate each axis of the robot by manual guidance operation to check for any abnormality. If any abnormality or unusual noise is found, stop the robot operation immediately and contact us.

* This function is available only in vertical articulated (6-axis) robot systems.

* Models and controller system versions in which this function is available:

Model : TV800

Parameter : FHT: V14.07 or later

Main : X8GCAV-14I or higher (vertical articulated (6-axis) robot system)

Servo : X8GLA-14C (TS3000 series), X8YCB-14C (TSL3000 series)

[U55] TOOL CALIBRATION FUNCTION

Tool calibration function

[U55] TOOL CALIBRATION FUNCTION
{Table 1}
{0:Disable,1:Enable}
= 0
{tool name}
= “TL001”
{original tool offset}
= 0.000 0.000 0.000 0.000 0.000 0.000
{reference point}
= 0.000 0.000 0.000 0.000 0.000 0.000
{correction tool offset}
= 0.000 0.000 0.000 0.000 0.000 0.000
{Table 2}
•
•
•
{Table 8}
{0:Disable,1:Enable}
= 0
{tool name}
= “TL008”
{original tool offset}
= 0.000 0.000 0.000 0.000 0.000 0.000
{reference point}
= 0.000 0.000 0.000 0.000 0.000 0.000
{correction tool offset}
= 0.000 0.000 0.000 0.000 0.000 0.000

{Table 1} to {Table 8}

{0: Disable, 1: Enable}

Setting: **= (Enabling/disabling function)**
Data format: Integer type
Data units:
Data range: 0 to 1
Data count: 1
Description: This parameter enables/disables the tool calibration function. For details of the tool calibration function, refer to the "Operator's Manual (vertical articulated system)".
0: Function disabled
1: Function enabled

{tool name}

Setting: **= "(Tool coordinate name)"**
Data format: Character string
Data units:
Data range:
Data count: 1
Description: Set the name of the tool coordinate to be calibrated. An array type can also be set.

{original tool offset}

Setting: **= (Tool coordinate value (X axis)) (Tool coordinate value (Y axis))**
(Tool coordinate value (Z axis)) (Tool coordinate value (A axis))
(Tool coordinate value (B axis)) (Tool coordinate value (C axis))
Data format: Real number type
Data units: mm or deg
Data range:
Data count: 6
Description: Reference tool coordinate values

{reference point}

Setting:	**= (MI/F coordinate value (X axis)) (MI/F coordinate value (Y axis))**
	(MI/F coordinate value (Z axis)) (MI/F coordinate value (A axis))
	(MI/F coordinate value (B axis)) (MI/F coordinate value (C axis))
Data format:	Real number type
Data units:	mm or deg
Data range:	
Data count:	6
Description:	Reference MI/F (mechanical interface) coordinate values

{correction tool offset}

Setting:	**= (Tool coordinate calibration value (X axis))**
	(Tool coordinate calibration value (Y axis))
	(Tool coordinate calibration value (Z axis))
	(Tool coordinate calibration value (A axis))
	(Tool coordinate calibration value (B axis))
	(Tool coordinate calibration value (C axis))
Data format:	Real number type
Data units:	mm or deg
Data range:	
Data count:	6
Description:	Tool coordinate calibration values

*This function is available only in vertical articulated (6-axis) robot systems.

*Controller system versions in which this function is available:

Main : X8GCAV-14I or higher (vertical articulated (6-axis) robot system)

Tool coordinate names and values are updated by operations on the utility screen.

[U56] FUNCTION SELECT SWITCH2

Controller function enabling/disabling selection

[U56] FUNCTION SELECT SWITCH2
= 0 0 0 0 0 0
= 0 0 0 0 0 0
= 0 0 0 0 0 0
= 0 0 0 0 0 0

Setting: **= (FUNCTION 1) (FUNCTION 2) (FUNCTION 3) (FUNCTION 4) (FUNCTION 5) (FUNCTION 6)**
= (FUNCTION 7) (FUNCTION 8) (FUNCTION 9) (FUNCTION10) (FUNCTION11) (FUNCTION12)
= (FUNCTION13) (FUNCTION14) (FUNCTION15) (FUNCTION16) (FUNCTION17) (FUNCTION18)
= (FUNCTION19) (FUNCTION20) (FUNCTION21) (FUNCTION22) (FUNCTION23) (FUNCTION24)

Data format: Integer type

Data units:

Data range: 0 to 2

Data count: 6*4

Description: This selection enables/disables a controller function.
To each element, a function which can be enabled is assigned.
Basically, 1: Function ON, 0: Function OFF

The functions corresponding to each element are listed below.

Function	Item Name	Description	Setting Example	Supported Systems
(FUNCTION 1)	Servo alarm reset count change	Sets the number of alarm resets required to reset a servo alarm. 0 : 1 reset 1 : 2 resets (compatible with before)	[U56]Function switch2 = **1** 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA 6-axis
(FUNCTION 2)	Torque limit reset at servo ON	Torque limit reset at servo ON Resets the torque limit value set in the SCOL language when the servo is turned on. 0: Function disabled (not reset at servo on) (as before) 1: Function enabled (reset at servo on)	[U56]Function switch2 = 0 **1** 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA 6-axis
(FUNCTION 3)	Payload application method switching in combined operation	0: Compatible with conventional (applied only to operating reference axis) 1: Payload applied in consideration of each axis	[U56]Function switch2 = 0 0 **1** 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA 6-axis
(FUNCTION 4)	Non-procedural communication log save	Saves logs of non-procedural communications using IP1, IP2, and IP3. 0: Function disabled 1: Log files are saved in the controller. 2: Log files are saved in a USB memory inserted in the controller. *For details, refer to the "Communication Manual".	[U56]Function switch2 = 0 0 0 **1** 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA 6-axis
(FUNCTION 5)	PRINT binary output	Converts the two-digit hexadecimal notation enclosed by "<" and ">" in PRINT to COM1, IP1, and IP2 into binary and outputs it. e.g. PRINT IP1, "<0A>" -> Outputs LF 0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 **1** 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA 6-axis
(FUNCTION 6)	INPUT binary input	When two-digit hexadecimal notation enclosed by "<" and ">" is used in STRING.PAR, it is interpreted and handled as binary. 0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 **1** = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	SCARA 6-axis
(FUNCTION 7)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = **1** 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION 8)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 **1** 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	

Function	Item Name	Description	Setting Example	Supported Systems
(FUNCTION 9)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 **1** 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION10)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 **1** 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION11)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 **1** 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION12)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 **1** = 0 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION 13)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = **1** 0 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION 14)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 **1** 0 0 0 0 = 0 0 0 0 0 0	
(FUNCTION 15)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 **1** 0 0 0 = 0 0 0 0 0 0	
(FUNCTION 16)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 **1** 0 0 = 0 0 0 0 0 0	
(FUNCTION 17)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 **1** 0 = 0 0 0 0 0 0	

Function	Name	Description	Setting example	Available in
(FUNCTION 18)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 **1** = 0 0 0 0 0 0	
(FUNCTION 19)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = **1** 0 0 0 0 0	
(FUNCTION 20)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 **1** 0 0 0 0	
(FUNCTION 21)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 **1** 0 0 0	
(FUNCTION22)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 **1** 0 0	
(FUNCTION23)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 **1** 0	
(FUNCTION24)	Reserved	0: Disabled 1: Enabled	[U56]Function switch2 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 0 = 0 0 0 0 0 **1**	

EXTRNSEL.PAR Setting Program File for File Select Signal Line

Specify the program file corresponding to the file select signal line.

```
COPYRIGHT(C) 2020 BY SHIBAURA MACHINE CO.,LTD.
ALL RIGHTS RESERVED.
EXTERNAL SELECT FILE
*** [ 00 – 0F ] *****
= "PROG00"
= "PROG01"
= "PROG02"
= "PROG03"
= "PROG04"
= "PROG05"
= "PROG06"
= "PROG07"
= "PROG08"
= "PROG09"
= "PROG0A"
= "PROG0B"
= "PROG0C"
= "PROG0D"
= "PROG0E"
= "PROG0F"
*** [ 10 – 1F ] *****
= "PROG10"
= "PROG11"
= "PROG12"
= "PROG13"
```

The number of effective files differs with the bit length according to the setting of [U07], and the files are assigned to the file select signals in turn, starting with the top file.

Bit length 1 :	0	~	1 (binary number)	Two (2) files
Bit length 2 :	00	~	11 (binary number)	Four (4) files
Bit length 3 :	000	~	111 (binary number)	Eight (8) files
	.			
	.			
	.			
Bit length 8 :	00000000	~	11111111 (binary notation)	256 files

Specify the file name following the equal (=) code.
Unless the equal (=) code is specified at the top of the line, it is processed as a comment.

Set value (Example): **= "PROG1"** PROG1 is assigned to the file select signal.

Appendix 1. User Parameters Adoption History

Main / USER. PAR No.	TS3000 SCARA X8GCA **S**	TS3000 Cartesian Coordinate X8GCA **L**	TS3000 Wafer X8GCA **W**	TS3000 Vertical articulated X8GCA **V**	Details
[U00]~[U26]	01A	02C	02B	01A	Initial release
[U27]	02B	02C			STRINGIO.PAR response command set
				03A	Force detection limit (vertical articulated only)
[U28]	05F	12A			PASS operation mode switching function
				03A	Position correction amount limit (vertical articulated only)
[U29]	06A	12A		08C	Alarm history display switching function
[U30]	08D	*) 12A		*) 09A	3-axis origin position change function (SCARA only)
[U31]	08D	12A		09A	JOG speed change function
[U32]	08D	12A		09A	JOG speed
[U33]	08D	12A		09A	Inching speed
[U34]	08D	12A		09A	Inching amount
[U35]	08E	12A		09A	Selection of enabling or disabling sequence function *1) Fieldbus ladder switching function *2) Fieldbus data communication function
[U36]	*) 10A	*) 12A		10B	Vibration suppression function (available in TV800)
[U37]	10F	12A		11H	Alarm file dump function
[U38]	10G	12A		11H	Alarm in file operation via signals
[U39]	11A	12A		*) 11H → 13B	Override retention function in case of power failure *) Function implemented in 13B (vertical articulated)
[U40]	*) 11B	*) 12A		11H	SMOOTH speed control function (vertical articulated only)
[U41]	*) 11B	*) 12A		11H	Function to disable the SMOOTH alarm (vertical articulated only)
[U42]	11B	12A		*) 11I → 11K	Function to change INPUT error processing *) Function implemented in 11K (vertical articulated)
[U43]	*) 11C	*) 12A		11I	Function to disable configuration alarm of teach point movement (linear interpolating) (vertical articulated only)
[U44]	11C	12A		11K	Function to change signal point data name
[U45]	11D	12A		11K	Function to change output alarm signal
[U46]	11F	12A		11K	Speed pulse output function
[U47]	11G	12A		11M	PLC override 0 setting function

[U48]	*) 13D			12B	NCBOY additional axis function enabling/disabling selection
[U49]	*) 13D			12B	NCBOY additional axis inching amount setting
[U50]	13D			*) 13D → 13E	Signal feed hold function *) Function implemented in 13E (vertical articulated)
[U51]	13E			*) 13D	3-axis torque display change function (SCARA only)
[U52]	14F			*) 14D	Out of limit error disabling function
[U53]	*) 15G			15A	TORQUE command enabling/disabling setting (6-axis)
[U54]	*) 15G			14I	Collision detection function
[U55]	*) 15G			14I	Tool calibration function
[U56] FUNCTION1	16B			15C	Servo alarm reset count change
[U56] FUNCTION2	16B			15C	Servo ON torque limit reset function
[U56] FUNCTION3	16D			15C	Payload application method switching
[U56] FUNCTION4	16G			15G	Non-procedure communication log saving function

*) Only the parameter is added, and the function is not unimplemented.
*1) Sequencer : X8GCC-04A or higher (TS3000 series), X8YCC-04A or higher (TS3000 series)
*2) Main : X8GCAS-13G or higher (SCARA), X8GCAV-13D or higher (vertical articulated)
Sequencer: X8GCC-07B or higher (TS3000 series), X8YCC-07B or higher (TS3000 series)

Copyright 2020 by SHIBAURA MACHINE CO.,LTD.
All rights reserved.
No part of this document may be reproduced in any form without obtaining prior written permission from SHIBAURA MACHINE CO.,LTD.
The information contained in this manual is subject to change without prior notice to effect improvements.